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9. Chloe was investigating the effect of using catalysts on the rate of reaction.

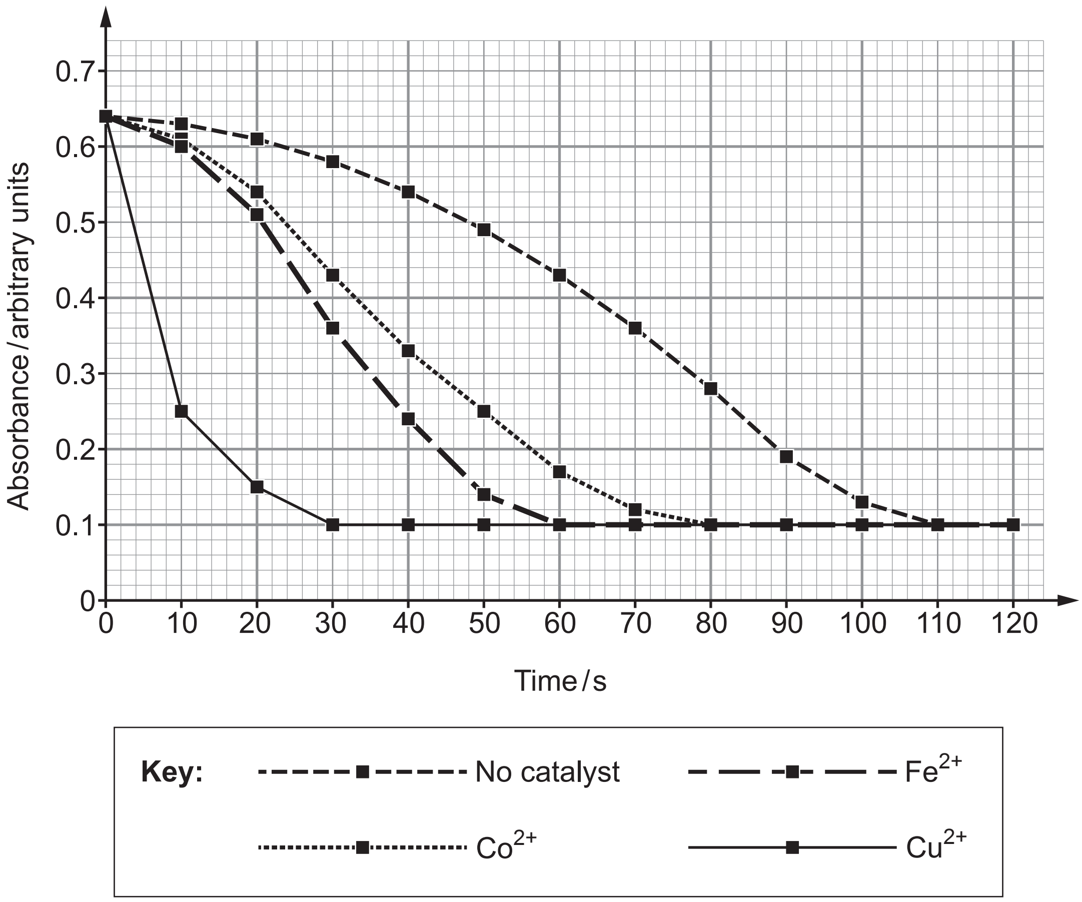
She added $50\,cm^3$ of $0.1\,mol\,dm^{-3}$ iron(III) nitrate solution to $50\,cm^3$ of $0.2\,mol\,dm^{-3}$ sodium thiosulfate solution. The reaction forms a deep violet iron(III) complex which is unstable and is gradually reduced to form a light green iron(II) complex.

Chloe monitored the rate of reaction by measuring the absorption of light at a wavelength of 500 nm every 10 seconds for two minutes using a data logger.

(a) The violet complex appears black at the beginning of the reaction. State the name of the technique used to monitor the rate of reaction by measuring the absorption of light. [1]

....

(b) Chloe repeated the experiment three times adding $1\,cm^3$ of a different catalyst each time at a concentration $0.10\,mol\,dm^{-3}$. Below is a graph showing her results:



(i) State which catalyst is the most effective. Explain your answer. [2]

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(ii) Calculate the initial rate of reaction for the reaction catalysed by the copper(II) ions. [2]

rate = ... s^{-1}

(iii) Each catalysed reaction contained the same number of moles of catalyst at the beginning of the reaction. Calculate the moles of catalyst left at the end of the reaction. [1]

moles = ... mol

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(c) Increasing temperature and the addition of a catalyst are two ways of increasing the rate of a reaction.

Using your knowledge of the Boltzmann distribution and particle theory, explain how the rate of reaction is increased using these two different methods.
You may use a diagram(s) to support your answer. [6 QER]

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